Ambient Systems

MVP Plan

Hardware Technology (IoT)

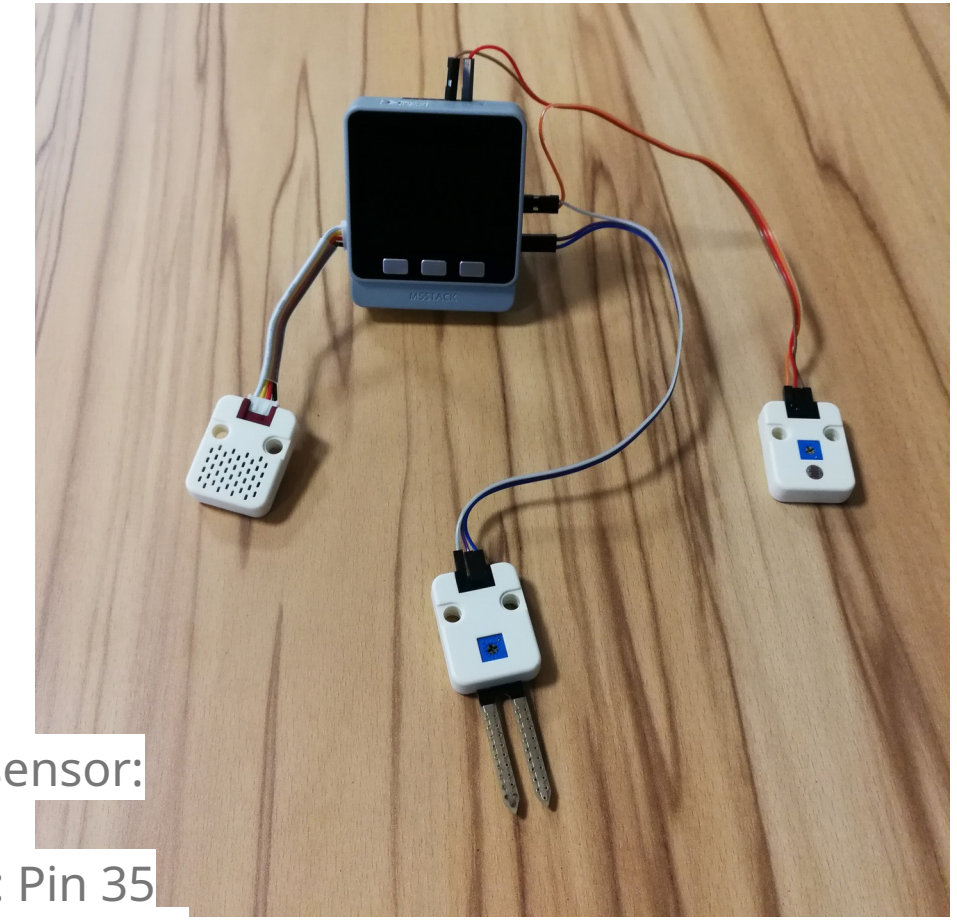
Overview

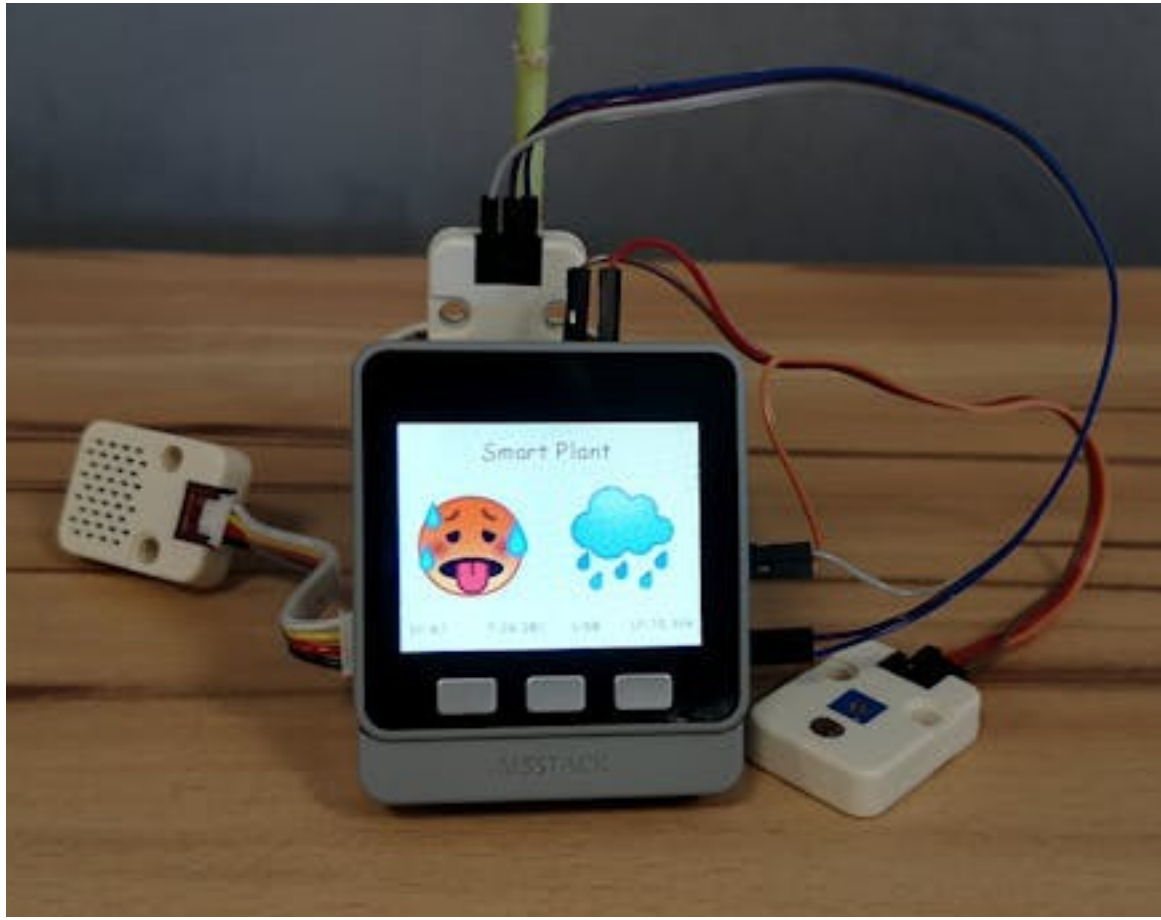
- m5Stack Demo
- Vertical Farming MVP
- Transportation
- Roadmap

m5Stack Demo

- M5Stack Core2 ESP32 IoT Development Kit for AWS IoT Kit
- Core2 for AWS repo
- CORE2 FOR AWS IOT KIT Guide
- Attachments:
  - Earth Moisture Sensor Unit Analog and Digital Output
  - CO2L Unit with Temperature and Humidity Sensor (SCD41)
  - Light Sensor Unit with Photo-resistance
- Device programming guide: UIFlow

Smart Plant Schematics





Smart Plant Schematics

Enviroment sensor: Port A

Earth sensor: Pin 35

Photo resistor: Pin 36

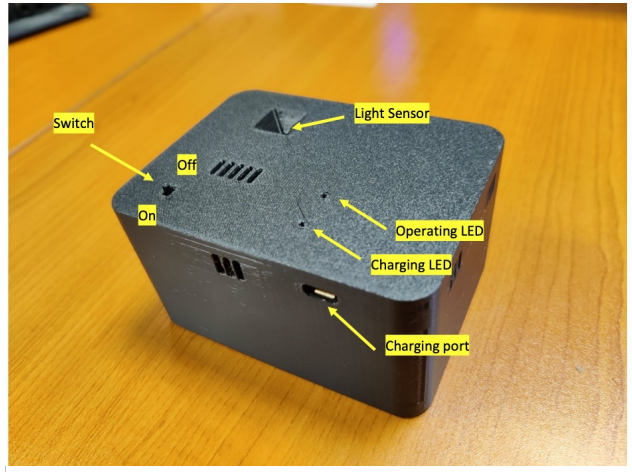
https://www.hackster.io/makerfactory_team/smart-plant-with-m5stack-89212e

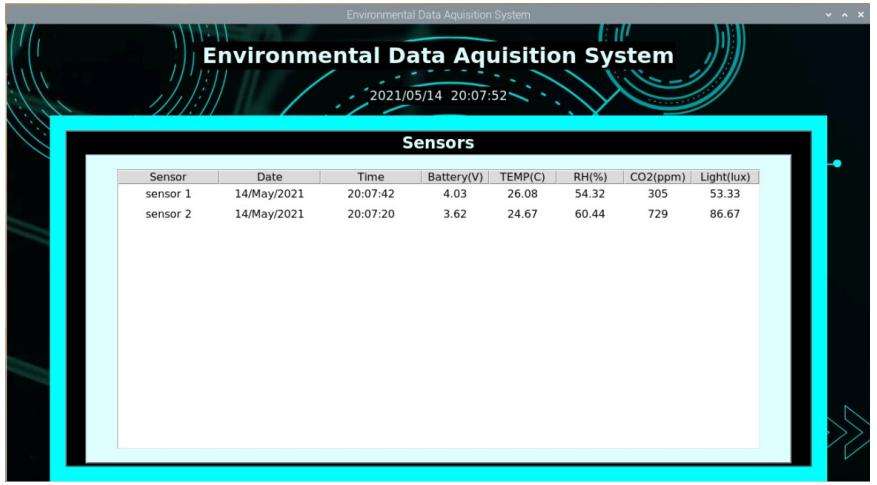
Attachments



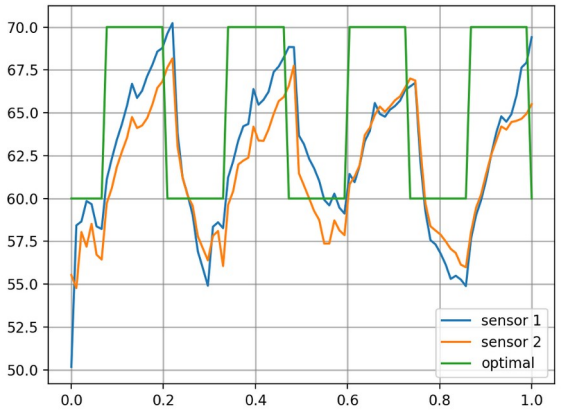


MVP #1: Vertical Farming







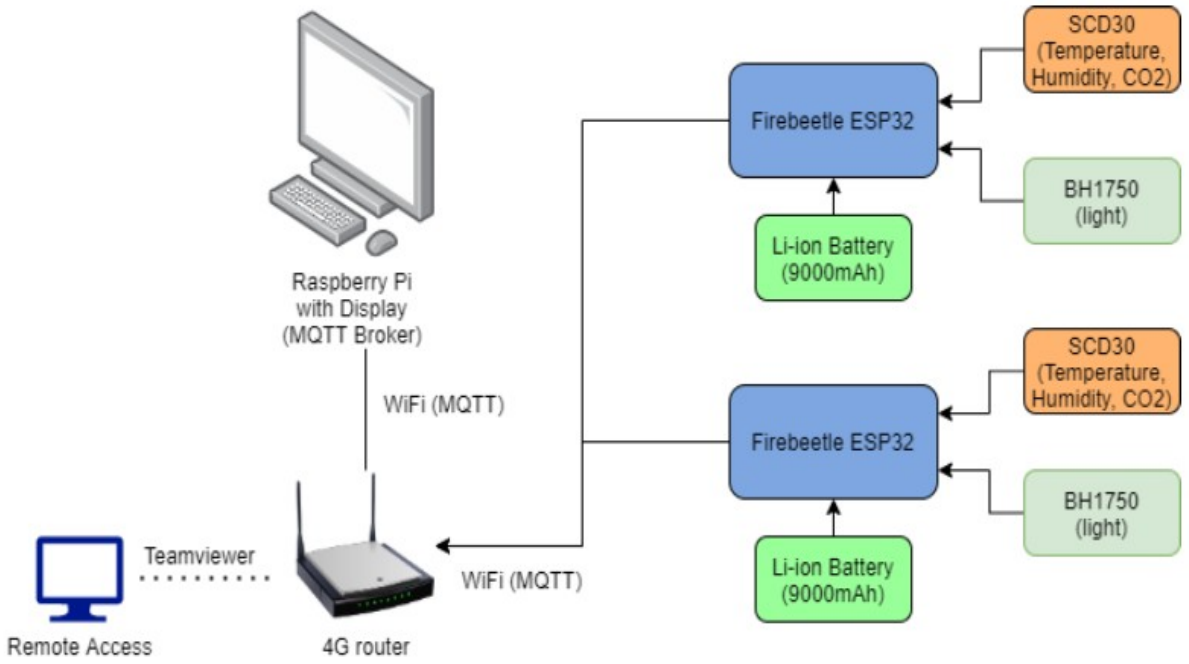




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Tech Category	Current State	Future Considerations	Technical Complexity
Sensing Device	• Firebeetle 32 ESP32: Microcontroller for data acquisition and pre-processing ( ). (-ion 9000mAh): Long-lasting power source for remote deployments. • Temp, Humi, CO2 sensor (Senserion SCD30): Environmental sensor gathering key data points. • Light Sensor (BH1750): Additional sensor for contextual data.	• Multimodal Sensing: additional sensors like soil moisture, air pressure, or audio for richer data collection. • Microcontroller Miniaturization: Smaller, more energy-efficient microcontrollers and sensors for compact deployments. • Sensor Fusion: Combine data from multiple sensors for e hed insights.	• Moderate – Requires knowledge of programming & electronics for customization. • Potential challenges include sensor calibration & accuracy maintenance, and data acquisition optimization for resource-constrained devices.
Connectivity	• 4G Router: Provides internet access for cloud communication and data transmission.	• Low-Power Wide-Area Networks (LPWAN): Long-range data transmission for remote locations. • Satellite communication: Global reach beyond cellular network coverage.	• Moderate – Requires knowledge of network protocols & security configurations. • Potential challenges include: ensuring reliable & secure data transmission in diverse environments and power optimization for extended connectivity in disconnected areas.
Processing Unit	• Raspberry Pi 400: Handles complex data analysis, decision-making, and user interface.	• On-device AI implementation: Real-time data analysis & intelligent actions without relying solely on cloud. • capabilities: Enhanced processing power & reduced reliance on cloud infrastructure.	• High – Requires expertise in programming, data analysis, & AI implementation. • Potential challenges include resource management & processing optimization for resource-constrained devices and balancing cloud dependence with on-device AI capabilities.
Display	• Computer Display: Visualizes data and insights for user interactions.	• Alternative display configurations: Consider touchscreens, e-paper displays, or even voice outputs. • Dynamic data visualization: Adapt visualizations based on user preferences & context.	• Moderate – Requires knowledge of display technologies & user interfaces. • Potential challenges include ensuring user-friendly & informative data visualization across different display options and optimizing power consumption for extended display operation.
Input	• Wireless Mouse: Allows user interaction with the system. • keyboard	• Alternative input configurations: Explore touch controls, voice commands, or gesture recognition.User experience optimization: Adapt input methods based on application & user preferences.	• Moderate – Requires knowledge of input technologies & user interfaces. • Potential challenges include designing intuitive & accessible input methods for diverse users & environments and balancing flexibility with power efficiency and security considerations.
Power	• Power adaptor (For Rpi, 5v, 3A): Powers the Raspberry Pi. storage for sensing devices.	• Energy efficient power supply to be explored.	• Moderate – Requires understanding of power electronics & energy management strategies. * Ensuring reliable & sustainable power supply for long-term deployments. * Balancing power consumption with system functionality & performance.

System Diagrams (Hardware)



MVP #3: Transportation (to be developed)

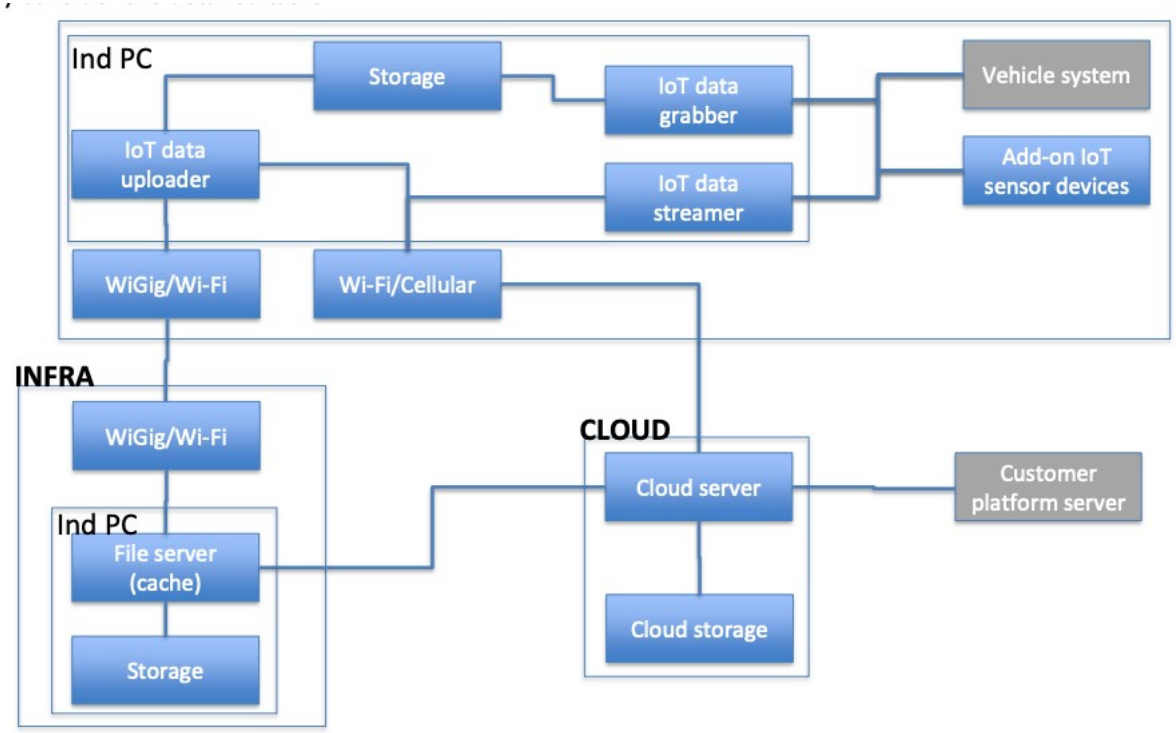
- Two units:
  - One connected to vehicle
  - One connected to ‘collection point’ with access to wi-fi and or data
- Data from vehicle:
  - Telematics
  - In-vehicle camera recordings (door-facing)
    -facing camera recordings
  - Operational and financial data (e.g., fuel)
  - In-vehicle camera recordings (safety and security)
  - Bus driver-facing and front-facing camera recordings
- Device Assembly
- Vehicle-Device Attachments

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Parts breakdown and cost estimate for MVP development

- IoT/data collection potential (TBC) equipment for MVP development (estimated total: **SGD 10229** excluding yellow items)
- Function: Driver recorder, in-vehicle, driver facing, etc
  - USB camera (connected to PC) x6 – E.g., C922n (**SGD 89** on amazon.sg)
  - (Optional) USB extension cable x6 – E.g., Pluggable 5m USB 2 active extension (**SGD 18** on amazon.sg)
- Function: (Optional) Geotagging
  - USB GPS/GNSS receiver (connected to PC) – E.g., (**SGD 371** on mouser.sg)
  - (Optional) USB extension cable – E.g., UGREEN Micro USB to USB 2.0 3m (**SGD 14** on amazon.sg)
- Function: Sensor data logging (e.g., accelerometer, IR)
  - USB accelerometer (connected to PC) – E.g., Yocto-3D (~**SGD 77** on yoctopruce.com)
  - USB IR sensor (connected to PC) – E.g., Yocto-Proximity (~**SGD 34** on yoctopruce.com)
  - USB extension cable x2 – E.g., Pluggable 5m USB 2 active extension (**SGD 18** on amazon.sg)
- Function: wireless data communication
  - Edge x1
    - mmWave communication unit (Edge) – (previously ~**SGD 750** from vendor)
    - 2.5 GbE Industrial PoE injector – (previously **SGD 291** from vendor)
    - Ethernet cable x2 – E.g., UGREEN cat 6 5m cable (**SGD 16** on amazon.sg)
    - (Tentative) DC power system (e.g. solar + battery) – TBD
    - (Optional) Internet connectivity (with cloud server or global IP) – TBD
  - OBU x1
    - mmWave communication unit (OBU + custom) – (previously ~**SGD 750 + TBC (estimate SGD 300)** from vendor)
    - 2.5 GbE PoE injector – E.g., TRENDnet (**SGD 63** on amazon.sg)
    - Ethernet cable x2 – E.g., UGREEN cat 6 5m cable (**SGD 16** on amazon.sg)
    - (Tentative) DC-AC power inverter – E.g., BESTEK 300W pure sine wave inverter (**SGD 78** on amazon.sg)
    - USB LTE/5G dongle (with data SIM card and plan) – E.g., Huawei 4G USB dongle (**SGD 78** on challenger.sg)
- Function: device and data handling
  - Industrial PC (OBU) – E.g, Vecow SPC-7100 with Intel Core i5, 16GB, 512GB – (previously **SGD 3706** from vendor)
  - Industrial PC (Edge) – E.g. Vecow ECX-3000 with Intel Core i5, 16GB, 512GB – (previously **SGD 2975** from vendor)

System Schematic



Roadmap

- Identify requirements for a pilot implementation based on customer interviews.
- Hardware design improvements consisting of multimodal sensing, miniaturization, sensor fusion, Low-Power Wide-Area Networks (LPWAN) adoption.
- Hardware improvements for on-device AI implementation, alternative displays, and energy-efficient power supply.
- Create API guide on how to connect Ambient Systems and established ERP, IoT, and telematics solutions for different use cases and clients.

  field pilot study or studies with finance teams in construction, commercial real estate, and tation companies in Hong Kong or GBA or SEA, demonstrating financial benefits of the comprehensive solution.

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- Fix any hardware integration issues identified during testing.